Forest Fire Simulation – Rust Course Project Proposal

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Project Topic

For my Rust course project, I plan to create a Forest Fire Simulation. This project will focus on emergent behavior, where simple rules can create interesting and sometimes unexpected results. The simulation will show how a fire can spread through a forest depending on the number of trees, their locations, and random events.

The main goal of this project is to practice programming in Rust while learning how simple interactions can affect a larger system.

Project Background

My Forest Fire Simulation project is based on the Drossel–Schwabl forest-fire model. This model is a type of cellular automaton where each cell follows simple rules based on its current state and neighboring cells. The model demonstrates how tree growth and fire spreading can create complex patterns through simple interactions.

I plan to implement a simplified version of this model using Rust. The simulation will demonstrate how tree growth, random fires, and fire spreading can create different patterns over time.

Although the simulation is based on an existing mathematical model, I will write my own Rust implementation and add features such as a terminal display, adjustable settings, and statistics.

Project Description

The program will simulate a forest using a two-dimensional grid. Each cell in the grid will represent one of three possible states:

- Empty: There is no tree in the cell.
- Tree: The cell contains a living tree.
- Burning: The tree is currently on fire.

The simulation will follow a few simple rules:

1. An empty cell has a small chance of growing a new tree.

2. A tree can catch fire if one of its neighboring cells is burning.
3. A tree also has a small random chance of catching fire without a nearby fire.
4. A burning tree will become an empty cell in the next simulation step.

These rules will repeat for multiple generations. Over time, the forest may grow, experience large fires, or develop areas with different numbers of trees.

The interesting part is that even though each cell follows simple rules, the entire forest can show more complicated behavior.

Expected Behavior and Results

I plan to display the simulation in the terminal using different characters to represent empty space, trees and burning trees.

For example:

- . represents an empty cell.
- T represents a living tree.
- F represents a burning tree.

The program will update the grid after each generation so users can observe how the forest changes over time.

I also want to display basic information, such as the number of living trees, burning trees, and empty cells.

Users will be able to adjust settings such as the size of the forest, the probability of tree growth, and the probability of a random fire starting.

By changing these settings, I hope to observe how different conditions affect the spread of fires and the overall forest.

Implementation Plan

I will write the program using Rust and organize the code into separate modules to make it easier to understand and maintain.

I plan to use structs and enums to represent the forest and the different cell states. I will also use vectors to store the grid and loops to update the simulation.

The project will include functions for creating the forest, updating cells, spreading fires, and displaying results.

I will write unit tests to check that the simulation rules work correctly. I will also use ‘cargo fmt’ and ‘cargo clippy’ to keep the code clean and follow Rust programming standards.

The program will use stable Rust and will not require unsafe code.

Possible Challenges

One challenge I expect is making sure the simulation updates correctly. For example, all cells should be updated based on the previous generation instead of changing other cells during the same step.

Another challenge will be working with random probabilities for tree growth and fire spreading. I will need to make sure these probabilities work as expected.

I may also have some difficulties organizing the Rust code into modules because I am still learning the language.

To handle these challenges, I plan to develop the project one feature at a time and test each part before moving to the next.

Project Goals

By the end of the quarter, I hope to have a working Forest Fire Simulation that demonstrates emergent behavior through simple rules.

I want the program to be easy to use and allow users to experiment with different forest conditions.

Most importantly, I want to improve my understanding of Rust, especially structs, enums, vectors, modules, and unit testing.

I will document the project in a README file, including how to run the program, what worked, what challenges I faced, and what I learned during development.

References

Wikipedia contributors. (n.d.). *Forest-fire model*. Wikipedia. https://en.wikipedia.org/wiki/Forest-fire_model